

ADVANCED AI, DEEP GENERATIVE MODELS & AUTONOMOUS DECISION SYSTEMS

New Egyptian Baccalaureate (EG-Bac) Advanced AI Syllabus

ACADEMIC YEAR: 2024/2025 | GRADE 12 (EG-BAC ADVANCED TRACK)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Advanced Egyptian Baccalaureate textbook in Artificial Intelligence, Deep Generative Modeling & Autonomous Cognitive Systems: Bayesian networks, d-separation, Expectation-Maximization (EM) for GMMs, Variational Inference with Evidence Lower Bound (ELBO), VAE reparameterization trick, score-based diffusion models (DDPM), CLIP multimodal contrastive learning, Markov Decision Processes, Bellman optimality, Deep Q-Networks, Proximal Policy Optimization (PPO), DeepSpeed ZeRO distributed model training, INT8/FP4 quantization, and SHAP explainability.

CORE CURRICULUM HIGHLIGHTS:

- D-separation proofs, explaining-away dynamics, and variational lower bound (ELBO) derivations
- Diffusion reverse Markov chain sampling and CLIP multimodal cross-entropy alignment
- Bellman operator fixed-point convergence and PPO clipped surrogate objective proofs
- DeepSpeed ZeRO-3 memory partitioning, INT8 affine quantization, and Shapley feature attribution

APPROVED CHAPTER MODULES:

- Advanced Statistical Learning & Probabilistic Graphical Models	pp. 1 78 (3 core units)
- Deep Generative Models, Diffusion & Multimodal Architectures	pp. 79 162 (3 core units)
- Reinforcement Learning, Autonomous Agents & Decision Systems	pp. 163 242 (3 core units)
- Enterprise MLOps, Distributed Model Training & Responsible AI	pp. 243 324 (3 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Advanced Generative AI, Diffusion & RL Mathematical Compendium (8p - 22.1 KB)
Full Ministerial Textbook: 324 Pages - 31.2 MB (Direct on MOE Portal)
Publisher: Egyptian Ministry of Education & Technical Education
Official Portal: <https://moe.gov.eg/elearning/egbac/advanced-ai-g12> - Code: EGBAC-AI-G12
Legal Deposit: 2025/14892 - ISBN: 978-977-12-2793-27



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Advanced Egyptian Baccalaureate textbook in Artificial Intelligence, Deep Generative Modeling & Autonomous Cognitive Systems: Bayesian networks, d-separation, Expectation-Maximization (EM) for GMMs, Variational Inference with Evidence Lower Bound (ELBO), VAE reparameterization trick, score-based diffusion models (DDPM), CLIP multimodal contrastive learning, Markov Decision Processes, Bellman optimality, Deep Q-Networks, Proximal Policy Optimization (PPO), DeepSpeed ZeRO distributed model training, INT8/FP4 quantization, and SHAP explainability. D-separation proofs, explaining-away dynamics, and variational lower bound (ELBO) derivations Diffusion reverse Markov chain sampling and CLIP multimodal cross-entropy alignment Bellman operator fixed-point convergence and PPO clipped surrogate objective proofs DeepSpeed ZeRO-3 memory partitioning, INT8 affine quantization, and Shapley feature attribution

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Advanced Statistical Learning & Probabilistic Graphical Models	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Deep Generative Models, Diffusion & Multimodal Architectures	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Reinforcement Learning, Autonomous Agents & Decision Systems	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Enterprise MLOps, Distributed Model Training & Responsible AI	7 wks	15 pts	25.0%	3 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: ADVANCED STATISTICAL LEARNING & PROBABILISTIC GRAPHICAL MODELS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Advanced Statistical Learning & Probabilistic Graphical Models.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Advanced Statistical Learning & Probabilistic Graphical Models.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: DEEP GENERATIVE MODELS, DIFFUSION & MULTIMODAL ARCHITECTURES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Deep Generative Models, Diffusion & Multimodal Architectures.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Deep Generative Models, Diffusion & Multimodal Architectures.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: REINFORCEMENT LEARNING, AUTONOMOUS AGENTS & DECISION SYSTEMS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Reinforcement Learning, Autonomous Agents & Decision Systems.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Reinforcement Learning, Autonomous Agents & Decision Systems.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

ADVANCED AI, DEEP GENERATIVE MODELS & AUTONOMOUS DECISION SYSTEMS
Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion: $\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \partial \rho / \partial t = 0$
Normalized Standard Score: $Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES
Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary: $s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio: $\eta = (\text{Output}_{\text{actual}} / \text{Input}_{\text{standard}}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION

NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

Official Document Code: EGBAC-AI-G12

Curriculum Cycle: 2025/2026 Academic Year | National Edition

Official Portal: <https://moe.gov.eg/ar/elearning-content/>

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